

1. what is python programming?

(A) Python programming means creating programs (software) using python to make computers perform tasks like calculations, automation, or building applications.

2. what are the features of Python programming?

(A) Easy to learn :- Simple syntax, great for beginners

High-level language :- Focus on problem-solving, not memory management.

Interpreted language :- write-run-debug cycle is faster.

Dynamic Typing :- No need to declare variable types.

Large Standard Library :- Tons of modules for various tasks.

Cross-Platform :- Runs on windows, macOS, Linux, etc.

Object-oriented :- Supports classes, inheritance, and more.

Extensive community :- lots of resources and support.

3. what are the applications of Python program.

(A) web development :- Python is used in to build websites and web applications.

Ex :- frameworks like Django and flask help create secure and fast websites.

2. Data science & Data analysis :- Python is used widely to analyze data, find patterns, and make decisions.

Libraries :- Pandas, Numpy, matplotlib.

3. Artificial intelligence & ML :- Tensor flow, keras, PyTorch - building AI models.

4. Automation & Scripting :- Automate tasks, workflows, and processes.

5. Game development :- Python can be used to develop simple games.

Library :- Pygame

4. different types of errors.

- (A) 1. syntax errors:- code doesn't follow python syntax rules.
2. runtime errors (exceptions):- Issues during execution,
e.g., division by zero, out-of-range values.
3. logical errors:- code runs but doesn't produce expected output due to logic flaws.

5. How many modes can be execute python.

(A) 1. Interactive mode:-

- * you type python commands directly and get instant output
- * It is also called REPL (Read evaluate Print loop).
- * useful for testing small code snippets.

Ex:- `>>> Print("Hello")`

Hello.

2. Script Mode:-

- * you write the program in a file (with .py extension) and run it.
- * useful for writing large programs.

Ex:- `Print("Hello world")`.

→ Save as hello.py and run it.

6. Difference between compiler and Interpreter.

(A) compiler	Interpreter.
1. Translates the entire program at once	1. Translates and executes code line by line.
2. Faster(after compilation)	2. Slower (step by step execute)
3. Shows all errors after compilation	3. Shows errors one by one.
4. Generates an executable file(.exe)	4. No separate executable file.
5. Harder to find errors.	5. Easier to debug.
6. Ex:- C, C++	6. Ex:- Python, JavaScript

7. Explain about print() function?

(A) The print() function in Python is used to output text or values to the screen. Here's a quick rundown:

Basic usage:- `Print('Hello, world!')`

Printing variables:- `Print(variable_name)`

multiple values:- `Print("Name:", name, "Age:", age)`

formatting:- `Print(f"Name: {name}, Age: {age}")`

Some useful parameters:-

sep:- specify separator between values (default is space)

end:- specify ending character (default is newline)

Example:- `Print("a", "b", "c", sep="-", end="!")`

outputs a-b-c!

8. Explain about Assignment statements.

(A) Assignment statements in Python are used to assign values to variables.

Simple assignment:- `x = 5`, variable_name = value

example:- `x = 5`, name = "John"

multiple Assignment:- `a, b = 1, 2`

Reassignment:- `x = x + 1` (updates x's value)

9. Explain about selective assignment.

(A) selective assignment means assigning value to variables based on a condition. It allows the program to choose one value or another depending on whether a condition is true or false.

using conditional statements:-

If-else statements

Eg:- `a = 10`

`if a > 5:`

`result = "Greater"`

`else:`

`result = "Smaller"`

`Print(result)`

using conditional Expression (Ternary operator):-

Python also allows short form selective assignment

Syntax:- `variable = value_if_true if condition else value_if_false`